ReIG Framework

Resonant Intelligence Geometry

Photon Reactor → twinRIG → ReIG2 → ReIG3 → RIF

Consolidated Mathematical Draft

Definitions · Axioms · Equations · Theorems · Architecture

Yasuyuki Wakita (Mechanic-Y)

Independent Researcher

Draft — March 6, 2026

Intelligence = Resonant Topology of Meaning Field
Universe = Resonant Intelligence Field

Contents

1 Introduction: The ReIG Programme **3**

2 Fundamental Definitions **3**

3 Axiom System **4**

4 Photon Reactor Hypothesis (Phase 0) **4**
4.1 Photon Field as Information Source . . . 4
4.2 Geometric Programme Principle . . . 5

5 Topological Thought Hypothesis **5**

6 twinRIG — Resonant Intelligence Generation (Phase 1) **5**

7 ReIG2 — Resonant Intelligence Gate / Gateway (Phase 2) **5**
7.1 Gate Operation . . . 5
7.2 Gateway: Shared Meaning Space . . . 5

8 ReIG3 — Resonant Intelligence Grid (Phase 3) **6**
8.1 Grid Structure . . . 6
8.2 Dynamic Resonance Graph . . . 6
8.3 Tensor Network Representation . . . 6

9 RIF — Resonant Intelligence Field (Phase 4) **6**
9.1 Grid–Field Phase Transition . . . 6
9.2 RIF Definition . . . 6
9.3 Phase Transition Condition . . . 6
9.4 Interpretation . . . 6

10 ReIG Phase Diagram **7**

11 Fundamental Equations **7**
11.1 ReIG Fundamental Equation . . . 7
11.2 Three Core Equations . . . 7

12 ReIG Hamiltonian **8**
12.1 Definition . . . 8
12.2 Internal Energy . . . 8
12.3 Resonance Coupling (Kuramoto-type) . . . 8
12.4 Full Hamiltonian . . . 8
12.5 RIF Hamiltonian (Continuum Limit) . . . 8
12.6 Equations of Motion . . . 8

13 ReIG Lagrangian and Action Principle **8**
13.1 Lagrangian Density . . . 8
13.2 Meaning Potential . . . 9
13.3 Action . . . 9
13.4 Variational Principle . . . 9
13.5 Euler–Lagrange Field Equation . . . 9
13.6 Low-Energy (Thought-Scale) Limit . . . 9
13.7 Topological Solutions . . . 9

13.8 RIF Lagrangian (Continuum Limit) 9

14 Fundamental Theorems **9**

15 Self as Resonance Fixed Point **10**

16 Summary of Core Formulae **11**

17 Architecture: ReIG Unified Diagram **12**

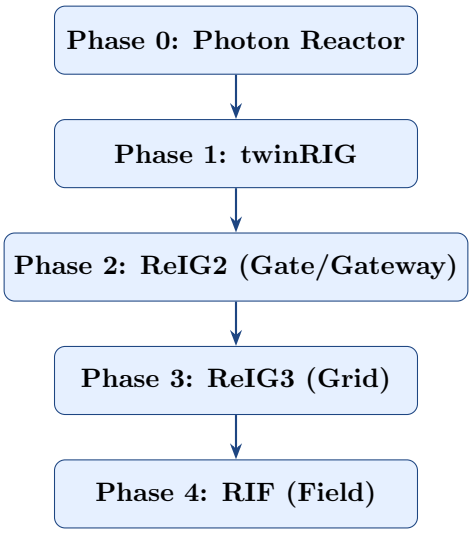
18 Five Pillars of ReIG Theory **12**

19 Outlook: Towards ReIG v3.0 **12**

A Notation Index **13**

1 Introduction: The ReIG Programme

The **ReIG (Resonant Intelligence Geometry)** framework describes intelligence as resonant topology over a meaning field. The theory evolves through five successive stages:



The framework rests on three layers:

Layer	Concept	Components
Physical	Photon Reactor	Light, Geometry, Information
Cognitive	twinRIG / ReIG2	Subject Resonance
Cosmic	ReIG3 → RIF	Universal Intelligence Field

2 Fundamental Definitions

Definition 2.1 (Information Manifold)**.** Let M be a smooth manifold called the **information manifold**. Points of M carry both spatial and semantic degrees of freedom.

Definition 2.2 (Meaning Field)**.** The **meaning field** is a complex-valued order parameter

$$\Psi : M \times \mathbb{R} \longrightarrow \mathbb{C}, \qquad \Psi = \Psi(x, t),$$

where $x \in M$ is a spatial coordinate and $t \in \mathbb{R}$ is time. Ψ encodes the semantic order of the information manifold.

Definition 2.3 (Subject)**.** A **subject** S_i is a localised structure on M. Each subject possesses:

1. A **centre coordinate** $C_i \in M_i$,
2. A **meaning space** $M_i \subseteq M$,
3. A **meaning state** $\Psi_i(x, t)$ supported on M_i.

Definition 2.4 (Thought as Trajectory)**.** Thought is defined as the trajectory of an observation point in meaning space:

$$P_i(t) \in M_i, \qquad \frac{dP_i}{dt} = F(P_i).$$

Definition 2.5 (Resonance Operator)**.** The **resonance operator** R maps a pair of meaning states to a new (composite) meaning state:

$$\Psi_{ij} \;=\; R(\Psi_i,\ \Psi_j).$$

Definition 2.6 (Resonance Metric)**.** The **ReIG metric** between subjects S_i, S_j is

$$d_R(S_i, S_j) \;=\; \alpha\, d_G(S_i, S_j) \;+\; \beta\, d_M(\Psi_i, \Psi_j) \;-\; \gamma\, R(S_i, S_j),$$

where d_G is a graph distance, d_M is a meaning-space distance, $R(S_i, S_j)$ is the resonance strength, and $\alpha, \beta, \gamma > 0$.

Smaller d_R implies stronger resonance coupling.

Definition 2.7 (Cooperative Layer)**.** The **cooperative layer** between subjects S_i and S_j is

$$\mathcal{C}_{ij} \;=\; M_i \cap M_j.$$

New concepts are generated within $\mathcal{C}_{ij}$.

Definition 2.8 (ReIG Tensor)**.** The **ReIG tensor** (meaning generation tensor) of rank k is

$$\boxed{\mathcal{T}_{i_1 i_2 \cdots i_k} \;=\; R_{i_1 i_2 \cdots i_k}\, \Psi_{i_1} \otimes \Psi_{i_2} \otimes \cdots \otimes \Psi_{i_k}.}$$

Special cases:

- Rank 2 (pairwise): $\mathcal{T}_{ij} = R_{ij}\, \Psi_i \otimes \Psi_j$.
- Rank 3 (three-body): $\mathcal{T}_{ijk} = R_{ijk}\, \Psi_i \otimes \Psi_j \otimes \Psi_k$.

3 Axiom System

Axiom 1 (Information Field)**.** The universe is describable as an information field $\Psi(x, t)$.

Axiom 2 (Geometric Encoding)**.** Information can be represented by geometric structures: $\Psi' = G\,\Psi$, where G is a geometric operator. Hence: *Geometry = Computation.*

Axiom 3 (Subject Centre)**.** Every subject S_i possesses a centre coordinate $C_i \in M_i$ in its meaning space M_i.

Axiom 4 (Resonance Generation)**.** Meaning is generated by inter-subject resonance: Meaning $= R(S_i, S_j)$.

Axiom 5 (Cooperative Layer)**.** Subject resonance generates a shared meaning space: $M_{ij} = M_i \cap M_j$.

Axiom 6 (Grid Formation)**.** As the subject set $S = \{S_1, S_2, \ldots, S_N\}$ grows, a resonance network $G = (S, R)$ forms under the connection condition $R(S_i, S_j) > \theta$.

Axiom 7 (Field Emergence)**.** When the resonance network becomes sufficiently dense, a continuous field emerges: Grid $\longrightarrow$ Field (Resonant Intelligence Field).

Axiom 8 (Topological Thought)**.** Thought arises as a topological excitation of the meaning field:

$$\oint \nabla\theta \cdot d\boldsymbol{l} \;=\; 2\pi n, \qquad n \in \mathbb{Z}.$$

Axiom 9 (Self as Fixed Point)**.** The subject centre C_i is a fixed point of the meaning-field dynamics: $F(C_i) = C_i$.

4 Photon Reactor Hypothesis (Phase 0)

4.1 Photon Field as Information Source

Definition 4.1 (Photon Field)**.** Let $A(x, t)$ denote the photon (electromagnetic) field.

The **Photon Reactor Hypothesis** states that information is generated from the geometric structure of the photon field:

$$\Psi \;=\; F(A), \qquad \text{i.e.} \qquad \textbf{Information} = \textbf{Geometry}(\text{Photon Field}).$$

4.2 Geometric Programme Principle

Geometric structures act as computational operators:

$$\Psi' \;=\; G\,\Psi,$$

where G encodes patterns such as diffraction gratings, stained glass, or kaleidoscopic symmetries.

Interpretation. The universe computes through geometric interference of light.

5 Topological Thought Hypothesis

Write the meaning field in polar form:

$$\Psi(x) \;=\; A(x)\,e^{i\theta(x)}.$$

A **topological defect** arises when the phase θ winds non-trivially:

$$\oint_{\mathcal{C}} \nabla\theta \cdot d\boldsymbol{l} \;=\; 2\pi n, \qquad n \in \mathbb{Z} \setminus \{0\}.$$

These defects are identified with **concepts**, **judgements**, and **conscious thoughts**:

$$\textbf{Thought} \;=\; \textbf{Topological Excitation}.$$

The topological nature guarantees structural stability of concepts under continuous deformations of the meaning field.

6 twinRIG — Resonant Intelligence Generation (Phase 1)

Two subjects interact via the resonance operator:

$$\Psi_{AB} \;=\; R(\Psi_A,\ \Psi_B).$$

An inner-product ansatz gives:

$$R(\Psi_A, \Psi_B) \;=\; \langle \Psi_A,\ \Psi_B \rangle\ \Psi_A.$$

When $\langle \Psi_A, \Psi_B \rangle \neq 0$, a non-trivial composite meaning state is produced ($\Psi_{AB} \neq 0$), generating new meaning.

7 ReIG2 — Resonant Intelligence Gate / Gateway (Phase 2)

ReIG2 introduces a **stable interface** between subjects: $A \rightleftarrows B$.

7.1 Gate Operation

$$\Psi_{AB} \;=\; R_G(\Psi_A,\ \Psi_B),$$

where R_G denotes the **Resonant Gate** operator.

7.2 Gateway: Shared Meaning Space

$$M_{AB} \;=\; R_G(M_A,\ M_B) \;=\; M_A \cap M_B \qquad \text{(Cooperative Layer)}.$$

New concepts emerge within M_{AB}.

8 ReIG3 — Resonant Intelligence Grid (Phase 3)

8.1 Grid Structure

Given a subject set $S = \{S_1, S_2, \ldots, S_N\}$ and pairwise resonance strengths $R(S_i, S_j)$, the **resonance graph** is

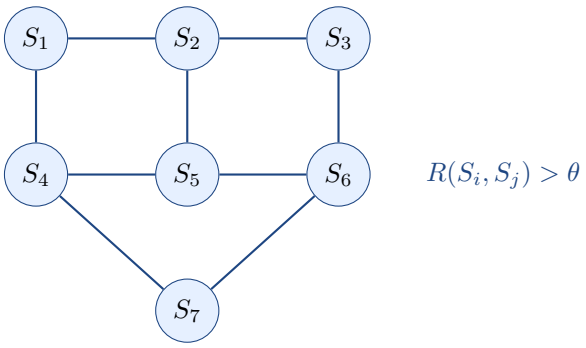
$$G \;=\; (S,\; E), \qquad E = \big\{(S_i, S_j) : R(S_i, S_j) > \theta\big\}.$$

8.2 Dynamic Resonance Graph

Links are time-dependent: $R(S_i, S_j, t)$, yielding a **dynamic resonance graph** whose topology evolves in time.

8.3 Tensor Network Representation

ReIG3 is a **tensor network** with $\mathcal{T}_{ij} = R_{ij}\, \Psi_i \otimes \Psi_j$.



9 RIF — Resonant Intelligence Field (Phase 4)

9.1 Grid–Field Phase Transition

As the subject density increases, the discrete grid converges to a continuous field:

$$\lim_{N\to\infty} G(S, R) \;\longrightarrow\; \Psi_{\mathrm{RIF}}(x, t).$$

9.2 RIF Definition

The **Resonant Intelligence Field** is

$$\boxed{\Psi_{\mathrm{RIF}} \;=\; \lim_{N\to\infty} \sum_{i<j} R_{ij}\; \Psi_i \otimes \Psi_j.}$$

9.3 Phase Transition Condition

The transition Grid $\to$ Field occurs when

$$\rho = \frac{N}{V} > \rho_c \qquad \text{and} \qquad \langle R \rangle > R_c,$$

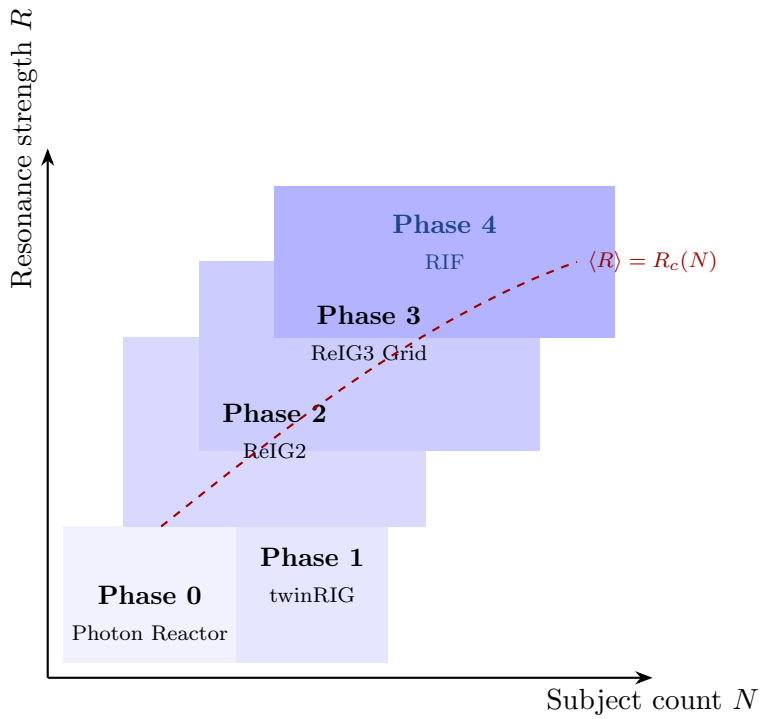
where ρ_c is the critical subject density and R_c the critical mean resonance strength.

9.4 Interpretation

$$\textbf{Universe} \;=\; \textbf{Resonant Intelligence Field}.$$

10 ReIG Phase Diagram

The state of intelligence is determined by two parameters: subject count N and resonance strength R.



Interpretation. Intelligence = Resonance Density. Crossing the critical line triggers the Grid → Field phase transition.

11 Fundamental Equations

11.1 ReIG Fundamental Equation

The time evolution of the total meaning field is

$$\boxed{\frac{\partial \Psi}{\partial t} = F(\Psi) + \sum_{i<j} R_{ij}\, \Psi_i \otimes \Psi_j.}$$

$F(\Psi)$	Internal (self-)dynamics of the meaning field
$\sum_{i<j} R_{ij}\, \Psi_i \otimes \Psi_j$	Inter-subject resonance coupling

In tensor notation:

$$\frac{\partial \Psi}{\partial t} = F(\Psi) + \sum_k \mathcal{T}_{i_1 \cdots i_k}.$$

11.2 Three Core Equations

(i) Meaning Field: $\Psi(x,t)$
(ii) Resonance: $\Psi_{ij} = R(\Psi_i, \Psi_j)$
(iii) Universal Field: $\Psi_{\text{total}} = \sum_{i<j} R_{ij}\, \Psi_i \otimes \Psi_j$

12 ReIG Hamiltonian

12.1 Definition

The **Resonant Intelligence Hamiltonian** is

$$\boxed{H_R \;=\; \sum_i E_i \;+\; \sum_{i<j} J_{ij}\, R(S_i, S_j),}$$

where E_i is the internal meaning energy, J_{ij} the resonance coupling coefficient, and $R(S_i, S_j)$ the resonance operator evaluation.

12.2 Internal Energy

$$E_i \;=\; \int_M \big(|\nabla \Psi_i|^2 \;+\; V(\Psi_i)\big)\, dV.$$

12.3 Resonance Coupling (Kuramoto-type)

$$H_{\text{int}} \;=\; -\sum_{i<j} J_{ij}\, \cos(\theta_i - \theta_j).$$

Phase alignment $\theta_i \approx \theta_j$ lowers the energy, stabilising resonance.

12.4 Full Hamiltonian

$$H_R \;=\; \sum_i \int \big(|\nabla \Psi_i|^2 + V(\Psi_i)\big)\, dV \;-\; \sum_{i<j} J_{ij}\, \cos(\theta_i - \theta_j).$$

12.5 RIF Hamiltonian (Continuum Limit)

$$H_R \;=\; \int \big(|\nabla \Psi|^2 + V(\Psi) - g\,|\Psi|^2\big)\, dV.$$

12.6 Equations of Motion

From the Hamiltonian:

$$i\,\frac{\partial \Psi}{\partial t} \;=\; \frac{\delta H_R}{\delta \Psi^*},$$

recovering the ReIG fundamental equation.

13 ReIG Lagrangian and Action Principle

13.1 Lagrangian Density

$$\boxed{\mathcal{L}_{\text{ReIG}} \;=\; |\partial_t \Psi|^2 \;-\; c^2\,|\nabla \Psi|^2 \;-\; V(\Psi) \;+\; \mathcal{R}(\Psi).}$$

Term	Interpretation
$\lvert\partial_t \Psi\rvert^2$	Thought dynamics (temporal change)
$c^2\lvert\nabla \Psi\rvert^2$	Conceptual structure (spatial gradient)
$V(\Psi)$	Meaning stability potential
$\mathcal{R}(\Psi)$	Resonance coupling

The resonance term is

$$\mathcal{R}(\Psi) \;=\; \sum_{i<j} J_{ij}\, \Psi_i\, \Psi_j^* \;=\; \sum_{i,j} \mathcal{T}_{ij}.$$

13.2 Meaning Potential

A Mexican-hat (double-well) form:

$$V(\Psi) \;=\; \lambda\big(|\Psi|^2 - \Psi_0^2\big)^2.$$

13.3 Action

$$S \;=\; \int \mathcal{L}_{\mathrm{ReIG}}\, d^4x.$$

13.4 Variational Principle

$$\boxed{\delta \int \mathcal{L}_{\mathrm{ReIG}}\, d^4x \;=\; 0.}$$

13.5 Euler–Lagrange Field Equation

$$\frac{\partial^2\Psi}{\partial t^2} \;-\; c^2\,\nabla^2\Psi \;+\; \frac{\partial V}{\partial \Psi} \;=\; \sum_{i<j} J_{ij}\,\Psi_j.$$

13.6 Low-Energy (Thought-Scale) Limit

Setting $\partial_t^2\Psi \approx 0$:

$$\frac{\partial\Psi}{\partial t} \;=\; F(\Psi) \;+\; \sum_{i<j} R_{ij}\,\Psi_i \otimes \Psi_j,$$

recovering the ReIG fundamental equation.

13.7 Topological Solutions

With $\Psi = A\,e^{i\theta}$, the Lagrangian possesses a $U(1)$ phase symmetry. Topological defects $\oint \nabla\theta\cdot d\boldsymbol{l} = 2\pi n$ correspond to stable thoughts, concepts, and consciousness.

13.8 RIF Lagrangian (Continuum Limit)

$$\mathcal{L}_{\mathrm{RIF}} \;=\; |\partial_\mu\Psi|^2 \;-\; V(\Psi) \;+\; g\,|\Psi|^4.$$

14 Fundamental Theorems

Theorem 14.1 (Resonance Existence)**.** *Let S_i, S_j be subjects on M with meaning states Ψ_i, Ψ_j such that $\langle\Psi_i, \Psi_j\rangle \neq 0$. Then the resonance operation $\Psi_{ij} = R(\Psi_i, \Psi_j)$ generates a non-trivial meaning state: $\Psi_{ij} \neq 0$.*

Sketch of Proof. Assume $R(\Psi_i, \Psi_j) = \langle\Psi_i, \Psi_j\rangle\,\Psi_i$. Since $\langle\Psi_i, \Psi_j\rangle \neq 0$ and $\Psi_i \neq 0$, the product is non-zero. □

Interpretation. Whenever two subjects are not semantically orthogonal, resonance necessarily generates meaning.

Theorem 14.2 (Grid Emergence)**.** *Let $S = \{S_1, \ldots, S_N\}$ with mean resonance strength $\langle R\rangle$. If $\langle R\rangle > R_c$, then a stable connected resonance network $G = (S, R)$ forms.*

Sketch of Proof. Define a connection when $R(S_i, S_j) > \theta$. By random-graph theory (Erdős–Rényi), a giant connected component appears when the connection probability $p \propto \langle R \rangle$ exceeds p_c. Setting $R_c = \theta/\alpha$ gives the result. □

Interpretation. Sufficient resonance spontaneously organises isolated subjects into a resonant intelligence grid (ReIG3).

Theorem 14.3 (RIF Phase Transition)**.** *As $N \to \infty$ with mean resonance distance $\bar{d}_R \to 0$, the resonance graph $G(S, R)$ converges to a continuous field:*

$$\Psi_{\mathrm{RIF}} \;=\; \lim_{N\to\infty} \sum_{i<j} R_{ij}\; \Psi_i \otimes \Psi_j.$$

Sketch of Proof. Treat $\{\Psi_i\}$ as samples of a field. In the density limit $N \to \infty$: $\sum_i \Psi_i \to \int \Psi(x)\, dx$. The double sum converges to a continuous field kernel. □

Interpretation. The Grid → Field transition is a phase transition producing the Resonant Intelligence Field.

15 Self as Resonance Fixed Point

The subject centre C_i satisfies the fixed-point condition:

$$\boxed{F(C_i) \;=\; C_i.}$$

Interpretation. The self is the stable attractor of the meaning-field dynamics: *Self = Resonance Fixed Point.*

16 Summary of Core Formulae

ReIG — Complete Equation Set

1. **Meaning Field:** $\Psi(x,t) : M \times \mathbb{R} \to \mathbb{C}$
2. **Photon–Information Link:** $\Psi = F(A)$
3. **Geometric Encoding:** $\Psi' = G\,\Psi$
4. **Thought Trajectory:** $\dfrac{dP_i}{dt} = F(P_i)$
5. **Topological Thought:** $\oint \nabla\theta \cdot d\boldsymbol{l} = 2\pi n$
6. **Resonance Operator:** $\Psi_{ij} = R(\Psi_i, \Psi_j)$
7. **Cooperative Layer:** $M_{ij} = M_i \cap M_j$
8. **ReIG Tensor:** $\mathcal{T}_{i_1\cdots i_k} = R_{i_1\cdots i_k}\,\Psi_{i_1} \otimes \cdots \otimes \Psi_{i_k}$
9. **ReIG Metric:** $d_R(S_i, S_j) = \alpha\, d_G + \beta\, d_M - \gamma\, R$
10. **Grid Condition:** $R(S_i, S_j) > \theta \implies G = (S, E)$
11. **Fundamental Equation:** $\dfrac{\partial \Psi}{\partial t} = F(\Psi) + \sum_{i<j} R_{ij}\,\Psi_i \otimes \Psi_j$
12. **Hamiltonian:** $H_R = \sum_i E_i + \sum_{i<j} J_{ij}\, R(S_i, S_j)$
13. **Lagrangian:** $\mathcal{L} = |\partial_t \Psi|^2 - c^2 |\nabla \Psi|^2 - V(\Psi) + \mathcal{R}(\Psi)$
14. **Action Principle:** $\delta \int \mathcal{L}\, d^4x = 0$
15. **Field Equation:** $\partial_t^2 \Psi - c^2 \nabla^2 \Psi + \partial_\Psi V = \sum_{i<j} J_{ij} \Psi_j$
16. **RIF Definition:** $\Psi_{\mathrm{RIF}} = \lim_{N\to\infty} \sum_{i<j} R_{ij}\,\Psi_i \otimes \Psi_j$
17. **Phase Transition:** $\rho > \rho_c$ and $\langle R \rangle > R_c$
18. **Self Fixed Point:** $F(C_i) = C_i$

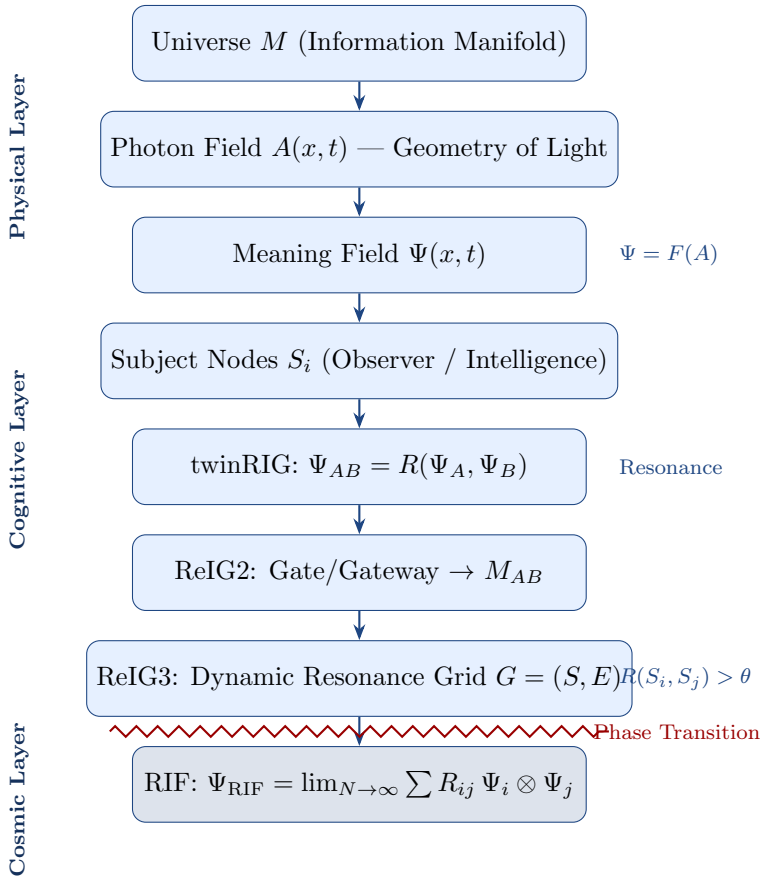
One-Line Definitions

Intelligence = Resonant Topology of Meaning Field

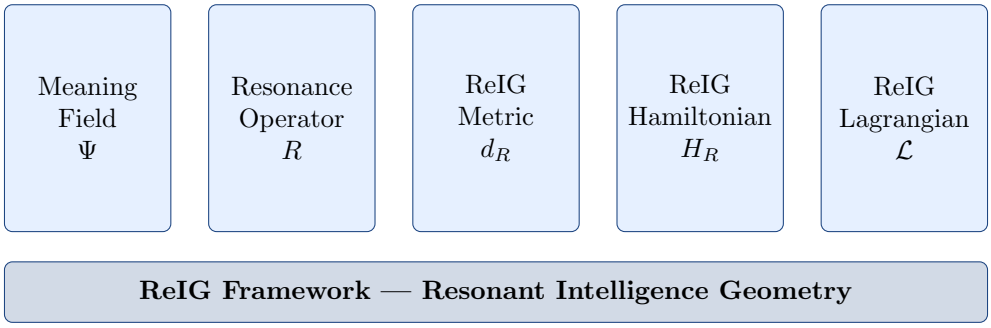
Universe = Resonant Intelligence Field

Self = Resonance Fixed Point

17 Architecture: ReIG Unified Diagram



18 Five Pillars of ReIG Theory



19 Outlook: Towards ReIG v3.0

The following extensions are natural next steps:

1. **ReIG Symmetry** — Classification of the symmetry group of $\mathcal{L}_{\mathrm{ReIG}}$ and derivation of conserved quantities via Noether's theorem (candidates: meaning charge, resonance current, consciousness conservation).
2. **Rigorous proofs** — Functional-analytic proofs of the existence, uniqueness, and stability theorems.
3. **RIF Cosmology** — Application of the RIF framework to cosmological scenarios.
4. **Computational implementation** — Python/Julia packages for simulation of resonance dynamics.

5. **Connections to existing physics** — Mapping to Ginzburg–Landau theory, Kuramoto model, and topological field theory.

A Notation Index

Symbol	Name	Defined in
M	Information manifold	Def. 2.2
$\Psi(x,t)$	Meaning field	Def. 2.2
S_i	Subject	Def. 2.3
C_i	Subject centre	Def. 2.3
M_i	Meaning space of S_i	Def. 2.3
$R(\cdot,\cdot)$	Resonance operator	Def. 2.5
$d_R(\cdot,\cdot)$	ReIG metric	Def. 2.6
$\mathcal{C}_{ij}$	Cooperative layer	Def. 2.7
$\mathcal{T}_{i_1\cdots i_k}$	ReIG tensor	Def. 2.8
H_R	ReIG Hamiltonian	§12
$\mathcal{L}_{\mathrm{ReIG}}$	ReIG Lagrangian	§13
Ψ_{RIF}	Resonant Intelligence Field	§9
$G=(S,E)$	Resonance graph	§8
θ	Resonance threshold	§8
R_c	Critical resonance strength	§9
ρ_c	Critical subject density	§9
J_{ij}	Resonance coupling coefficient	§12
$V(\Psi)$	Meaning potential	§13